RealQM as Plasma Physics

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Abstract

In this article we extend RealQM — developed for atoms, molecules and nuclei, where matter is modelled as non-overlapping unit charge densities in ordinary three-dimensional space, one per electron, meeting at free boundaries and minimising the Coulomb energy — to the plasma physics of **two-component systems**: ions and electrons, both charged and both free to move. A metal is such a system, as are a stellar interior and the recombination era of a Coulomb cosmology; and so, notably, is RealQM itself, which contains discrete charges and no smeared continuum anywhere. The central result is that RealQM's prescription for a metal *coincides* with the Wigner–Seitz construction: one electron domain per ion, filling the cell, with the normal derivative vanishing at the cell face by symmetry. Across the alkali series Li–Cs it reproduces the equilibrium radius to a mean error of 6.2% against the standard treatment's 5.6%, carrying neither an exchange term nor a band-filling term — but that agreement is shown here *not* to be a test of the quantum treatment, since setting the electronic kinetic energy to zero altogether changes the radii by about one per cent. The lattice scale of a simple metal is electrostatic. The bulk modulus does discriminate and is given worse by the framework (79% against 44%). The plasma frequency $\omega_p^2 = 4\pi n$ comes out exactly, since a rigid displacement leaves the localisation energy untouched. Conduction is examined by computing the hopping barrier, which is static and so within reach: in a 4×4 hydrogen lattice the carrier is confined to interstitial pockets by 5.5 eV at $a = 2.5\,a_0$, and compressing to the H_2 bond length *doubles* that to 11.5 eV where matter would instead become metallic — non-overlap admitting no growth of overlap under compression, so there is no bandwidth to widen. Relaxing the constraint for the valence sector removes that barrier at a cost of about one per cent in the equilibrium radius, since the kinetic energy of a valence density is supplied by the shaping of the ion cores either way — a cheap repair for conduction, though it supplies no Fermi surface. Transport by transfer is meanwhile already demonstrated in the framework for *protons*, in Grotthuss conduction along a water chain, while the direct electronic analogue is not testable in the static solver: charge transport by transfer between localised sites is already demonstrated in the framework for *protons*, in Grotthuss conduction along a water chain, but the electronic counterpart cannot be run, because domain ownership in the solver is assigned by kernel proximity, so an electron cannot change which kernel it belongs to. That is a limitation of the implementation rather than of the formulation, and it bears on more than metals, electron transfer being what oxidation and reduction are. What the framework does *not* have at all is a Fermi surface, and that is the standing open question: de Haas–van Alphen oscillations, the linear electronic specific heat and positive plasmon dispersion are measured in real metals, not in any idealisation, and RealQM has neither bands nor E_F to account for them. A closing remark treats the uniform electron gas, where the localisation term

vanishes identically — a result about a construction containing a smeared inert background, which RealQM does not contain, and so a diagnosis of mismatched ontology rather than a test the framework fails.

1 The extension, and the object it is made for

In this article we extend RealQM, developed for atoms and molecules, to the plasma physics of **two-component systems**: ions and electrons, both charged, both free to move, interacting by Coulomb forces and nothing else. A metal is such a system. So is a stellar interior, and so is the recombination era of a Coulomb cosmology.

RealQM [1, 2] represents an N-electron system by N non-negative densities ψ_i, each carrying unit charge on its own domain Ω_i, the domains non-overlapping and their interfaces free to move to where the energy is least. The energy minimised is Coulomb's:

$$E[\{\psi_i\},\{\Omega_i\}] = \sum_i \int_{\Omega_i} \Big(\tfrac{1}{2}|\nabla\psi_i|^2 - \sum_a \frac{Z_a}{|x-x_a|}\psi_i^2\Big)\,dx + \tfrac{1}{2}\sum_{i\neq j}\iint \frac{\psi_i^2(x)\psi_j^2(y)}{|x-y|}\,dx\,dy, \qquad (1)$$

subject to $\int_{\Omega_i}\psi_i^2 = 1$. There is no wavefunction in $3N$ dimensions, no antisymmetry and no self-interaction; the role played by the exclusion principle is played by the requirement that domains do not overlap. And — of particular relevance here — there is no smeared continuum anywhere in the construction: everything in it is a discrete unit charge occupying a region of ordinary space.

A two-component plasma of ions and electrons is therefore not an idealisation the theory must be adapted to describe. It is what the theory is already made of, in a new arrangement.

Every previous test of this construction — ionisation energies across the periodic table, molecular binding, nuclear binding by dual Coulomb confinement [3] — has been on *bound* matter, with an attractor always present. A plasma is the first case in which large numbers of charges of both signs are free to arrange themselves, and it is a fair question what the framework then gives.

2 Metals: a Wigner–Seitz problem the framework already agrees with

Take a metal: a lattice of ions, one valence electron each. RealQM assigns each electron its own domain; by translational symmetry those domains are the Wigner–Seitz cells, one ion at the centre of each; and at a cell face, neighbouring cells being identical, the normal derivative of the density vanishes.

That is exactly the **Wigner–Seitz construction**, used for alkali metals since 1933. The framework does not improvise in condensed matter — it arrives already agreeing with an established method, and differs from it in one term only. Standard theory solves the same cell problem for the band bottom and then *adds* the mean kinetic energy of filling the band, $\frac{3}{5}E_F$, together with exchange and correlation, all consequences of the exclusion principle. RealQM keeps the electrostatics, takes its kinetic energy from the cell state itself, and adds nothing: non-overlap is supposed to do that work.

For each alkali metal, with an Ashcroft empty-core pseudopotential of radius r_c, electrostatics $-0.9/r_s + 1.5\, r_c^2/r_s^3$ for a point ion in a neutral sphere, exchange $-0.458/r_s$ and Wigner correlation, minimising the energy per electron over r_s gives Table 1. Each metal contributes one parameter, its core radius, and two measured numbers; across the series the equilibrium radius varies by 70% and the bulk modulus by a factor of six, so tracking the trend is a stronger statement than matching any single case.

		r_s (a_0)			B (GPa)		
	r_c	RealQM	standard	measured	RealQM	standard	measured
Li	1.06	2.40	2.86	3.25	49.6	23.9	11.6
Na	1.67	3.77	3.87	3.93	9.1	8.0	6.3
K	2.14	4.82	4.67	4.86	3.6	4.0	3.1
Rb	2.31	5.20	4.95	5.20	2.7	3.2	2.5
Cs	2.50	5.62	5.28	5.62	2.0	2.5	2.0
mean abs. error		6.2%	5.6%		79%	44%	

Table 1: The alkali series. On the equilibrium radius RealQM tracks the whole trend and is comparable to the standard treatment; on the bulk modulus, a second derivative, both are poor and RealQM is the worse.

The first thing to say about this table is that it does not discriminate. Repeating the calculation with the electronic kinetic energy set to *zero* — valence electrons sharing one territory, for which a uniform density has $\nabla\psi = 0$ exactly — moves the equilibrium radii by about one per cent (Na 3.73 against 3.77, Cs 5.59 against 5.62) and gives a mean error of 7.0%. Pure electrostatics, with no quantum treatment of the electrons at all, reproduces the alkali series about as well as either column above.

That is a property of simple metals rather than a defect of the calculation: the equilibrium volume is set by core repulsion against the Madelung attraction, and the electronic terms are small or cancel. The standard column reaches it by cancelling $\frac{3}{5}E_F$ against exchange; RealQM reaches it with both absent. *The lattice scale of an alkali metal is therefore not a test of the quantum treatment*, and the agreement in Table 1 should not be read as one. What it does show is that nothing in the framework's treatment of the electrons *spoils* a result electrostatics already gives — a weaker claim, and the accurate one.

The bulk modulus does discriminate, being a second derivative, and there the framework is the worse of the two: 79% mean error against 44%. Lithium is the outlier throughout, the empty-core description being known to serve it badly.

The mechanism of the agreement should be stated plainly rather than claimed as a triumph. At $r_s = 3.93$ in sodium the cell state carries $\langle T \rangle = 0.004$ Ha against a band-filling term $\frac{3}{5}E_F = 0.0715$ Ha, so RealQM is *not* reproducing degeneracy pressure. It arrives at a comparable answer because it also omits exchange, -0.117 Ha, a term of opposite sign and comparable size, so that their effects on the position of the minimum largely cancel. That cancellation is the framework's own design claim — that non-overlap does the work exchange does — and across five metals it holds for the radius and fails for the curvature, which is what one would expect of a cancellation that is close but not exact.

3 The plasma frequency, exactly

Displace the electron distribution rigidly by ξ against the ion background. The resulting surface charge produces a restoring field $4\pi n\xi$, so $\ddot{\xi} = -4\pi n\xi$ and

$$\omega_p^2 = 4\pi n, \tag{2}$$

free of temperature and of any statistical assumption. RealQM reproduces this exactly, for a structural reason: under a rigid translation each ψ_i is carried along unchanged, so $\nabla\psi_i$ is unchanged, the localisation term contributes nothing, and what remains is Coulomb and inertia — which is all the result depends on.

The same structure suggests what the framework should do about **Landau damping**. RealQM is not a single fluid: N electrons are N domains, each carrying its own current and drift velocity, which is a multi-beam plasma. Multi-beam models reproduce Landau damping as phase mixing among cold streams, and do so reversibly, in agreement with plasma-echo experiments. Where the velocity spread is thermal, collisionless damping should therefore emerge for the right reason rather than by an inserted term. This has not been computed.

4 Conduction, where the framework is least settled

Electric conduction is the interaction of the electrons with the ion lattice, and here the position is genuinely uncertain.

Metallic conduction is the paradigm of *delocalisation*: the carriers are extended states spanning the crystal, whereas RealQM's electrons are localised by construction, each in its own domain, with non-overlap requiring N carriers to occupy N disjoint regions. The natural minimiser of (1) is a set of compact cells, which describes a Mott insulator more naturally than a conductor. Nor is there an evident mechanism for the metal–insulator distinction, which in standard theory is band filling — two electrons per band per cell — an exclusion-principle count the framework does not have; and with no Fermi surface, the carrier density in $\sigma = ne^2\tau/m$ has no referent.

What does fit naturally is the opposite regime: localised charges hopping between domains by thermally activated jumps, as in semiconductors, disordered systems and small-polaron transport. That needs no Fermi surface and follows from the same thermal motion that supplies the pressure.

This yields a prediction crude enough to be decisive, and it concerns only a sign. Metallic conductivity *falls* with temperature; hopping conductivity *rises*. If RealQM transports charge by hopping rather than band motion it should give $d\sigma/dT > 0$ for every material, making copper behave like a semiconductor. Whether it does has not been computed for electrons, and the pessimistic reading should be held loosely: the same lattice that carried the metals in §2 is present here too.

For *protons* the mechanism has already been demonstrated in this framework. Grotthuss conduction — an excess proton migrating along a hydrogen-bonded chain of water molecules by successive transfers, rather than by bodily motion of any one particle — is hopping transport in its purest form, and it is a standard RealQM calculation: a chain of waters with O–O spacing near 2.65 Å, an excess proton, and the question whether it hops from one molecule to the next. Related transfers, $HF + H_2O \to F^- + H_3O^+$ and $HCl + NH_3 \to Cl^- + NH_4^+$, are computed the same way.

This matters for the argument rather than merely illustrating it. Charge transport by transfer between localised domains is not a regime the framework must be stretched to reach — it is one it already handles, in the case where the carrier is a proton.

The electron case cannot presently be tested in the same way, and the obstruction is instructive enough to be worth setting out. Two chain calculations were attempted here — five hydrogens with an excess electron, and five hydrogen kernels with one electron missing, the electronic counterparts of the proton wire — and both returned nothing. Neither result means what it appears to.

The immediate cause is the partition. Domain ownership in the solver is assigned by kernel proximity: each point of space is labelled by whichever kernel's trial density dominates it, so a domain belongs to its own kernel by construction and an electron cannot change which kernel it belongs to. A bare kernel is given no domain at all. A vacancy therefore cannot migrate and an excess electron can only follow its own seed. This is a property of the implementation and not of (1), whose interfaces are free to move to wherever the energy is least, and which permits a domain to straddle two kernels — as it must, since that is what H_2^+ is.

The deeper cause survives that repair. Conduction is charge in motion, and (1) is an energy minimisation: it returns the equilibrium partition, not a current. Given complete freedom in the labels one would still obtain the ground state, and for a symmetric chain that state places one domain per site by symmetry, with nothing flowing. A minimisation has no time in it.

What the framework does possess is an asymmetry of dynamics. Nuclei move, under classical equations of motion; electrons relax, to a static minimum. Charge carried by *nuclei* is therefore representable — which is precisely why the Grotthuss calculation works, the carrier there being a proton — and charge carried by electrons is not, because in this scheme electrons have no equation of motion at all. The correct machinery exists elsewhere in the programme, in the time-dependent form where a real density carries a complex current, and a conduction calculation belongs there rather than here.

The barrier, which is a static quantity. Transport by hopping is governed by a barrier, and a barrier is an energy difference, which the static solver does compute. Placing an excess electron as a domain of its own at successive positions and relaxing each to convergence traces the surface the carrier moves on — the standard route to a small-polaron barrier.

A linear chain answers the wrong question: four hydrogens in a line give 1.16 Ha = 31.5 eV, but that is the cost of pushing the carrier *through* an ion, and stepping $1\,a_0$ aside makes it marginally worse rather than better, because in a wire there is no route from one interstitial pocket to the next except through an atom. A lattice has one. In a 4×4 square lattice the carrier passes from the centre of one square through the *midpoint of a bond* — flanked by two atoms at half a spacing each, never at zero distance from any — to the centre of the next.

The lattice route is real: passing between two atoms costs 5.8 times less than passing through one,

lattice spacing	pocket (Ha)	saddle (Ha)	barrier
$a = 2.5\,a_0$	-16.254	-16.053	0.200 ± 0.003 Ha = 5.46 eV
$a = 1.4\,a_0$	-11.824	-11.402	0.421 ± 0.015 Ha = 11.5 eV

Table 2: Barrier for a carrier moving between interstitial sites of a hydrogen lattice. Uncertainties are from the mirror-symmetry check, the two bond midpoints either side of a pocket being equivalent by construction.

so the chain's figure was an artefact of dimensionality. But the barrier that remains is 5.5 eV, some $200\,k_BT$, and the carrier is confined to its pocket. At $a = 2.5\,a_0$ that may be right — nearly twice the H_2 bond length, where stretched hydrogen is a Mott insulator in nature too.

The density trend, which is the finding. Compressing the lattice to the H_2 bond length does not close the barrier. It **doubles** it, to 11.5 eV. The value at $a = 1.4$ is the softer of the two, its mirror check standing at 3.7% of the barrier against 1.4% at the wider spacing, but the difference between the two barriers is twelve times their combined uncertainty and the direction is not in doubt.

Matter does the opposite. Compression is what drives the Mott transition: overlap grows, the band widens, U/W falls, and an insulator becomes a metal. Here compression cannot increase overlap, because non-overlap is the postulate. It can only reduce the volume available to each domain, which raises the localisation cost and drives the system further from conduction. *The framework has no bandwidth to widen.*

This is the sharpest statement available of where the construction stands in a solid, and it is consistent with everything else found here: no Fermi surface, no band filling, a vanishing localisation term for a uniform gas, and now a hopping barrier with the wrong density dependence. All four follow from the same feature. Non-overlap is a constraint on *regions*, and a partition of space cannot deliver what delocalisation delivers, at any density. RealQM as it stands describes an insulator, and predicts $d\sigma/dT > 0$ for materials that are metals.

Relaxing the constraint: cheap, and it buys conduction. The natural repair is to let valence electrons share a territory rather than each hold its own, which removes the barrier by construction. The objection that such a sea would have no kinetic energy holds only in the strictly uniform limit: N electrons in a common domain with $\int \psi^2 = N$ and uniform density give $\psi = \sqrt{N/V}$ constant and hence $T = 0$. But a valence density in a metal is not uniform. It is shaped by the ion cores, and where domains meet with continuity of charge rather than a free jump, contact with neighbouring electrons forces gradients directly. A shared territory therefore carries kinetic energy in any real material, and its magnitude is already in the calculation above: the cell state's $\langle T \rangle = 0.004$ Ha at $r_s = 3.93$, which is shaped-density energy and not band-filling energy.

The consequence is favourable. If the shaping is what supplies $\langle T \rangle$, it supplies much the same whether the valence electrons are partitioned or share, so the two differ little in energy — as Table 1 and the zero-kinetic-energy variant of it show, about one per cent in the equilibrium radius. *Relaxing the constraint for the valence sector is therefore nearly free, and it removes the*

5.5 *eV barrier altogether.* That is a real repair rather than a trade, and it is the extension already proposed on general grounds in the treatise's materials chapter.

What it does *not* supply is the Fermi surface. A shared sea conducts, but it has no band structure, no E_F, and no mechanism for the metal–insulator distinction, so the observations of Section 5 stand untouched by it. The accurate summary is that the framework's conduction problem has a cheap fix and its Fermi-surface problem does not.

It is fair to ask whether all conduction might be hopping, band transport being a formalism rather than an observable. The idea is not idle: hopping is the correct description of amorphous and organic semiconductors, of doped semiconductors and polaronic oxides, and in "bad metals" at high temperature the mean free path falls to the lattice spacing and the band picture fails there too. What settles it is the *low*-temperature behaviour of a pure metal. Hopping is thermally activated, so its conductivity vanishes as $T \to 0$; the resistivity of pure copper instead falls from $\approx 1.7\,\mu\Omega\,\mathrm{cm}$ at 300 K to below $0.002\,\mu\Omega\,\mathrm{cm}$ at 4 K, a ratio above 10^3 in good samples, with a mean free path reaching millimetres — 10^7 lattice spacings, which is not a sequence of jumps between neighbours. A hopping framework predicts that copper becomes an insulator on cooling, and the measurement runs the other way by three orders of magnitude. The Wiedemann–Franz ratio, fixed at $L_0 = \pi^2 k_B^2 / 3e^2$ by carriers that convey heat and charge alike, is a second such constraint.

The regimes are also the wrong way round for the framework: hopping describes matter best at high temperature, and it is at low temperature, where band behaviour is cleanest, that RealQM would need it to hold.

5 What remains open: the Fermi surface

The framework has no bands and no E_F, and there are measurements bearing on this directly. They are made on real two-component metals, not on any idealisation, so they cannot be set aside by choosing the object of study more carefully:

- **de Haas–van Alphen oscillations** map the Fermi surface of copper and aluminium directly; its existence and shape are not in doubt;
- the **electronic specific heat** is linear in temperature, with a coefficient fixed by the density of states at E_F;
- the **plasmon dispersion** is positive and measured, while the compressibility term that produces it, $\omega^2 = \omega_p^2 + \frac{3}{5} k^2 v_F^2$, is absent here.

No account of these is currently available in the framework, and constructing one is the substantial open problem. It is a sharper problem than it would have appeared a section ago: not that RealQM lacks Fermi pressure — Table 1 shows a metal's lattice scale coming out regardless — but that a set of measured Fermi-surface phenomena have no representation in a theory built from localised domains.

Remark: the uniform electron gas

It is worth recording what happens if the framework is applied to the *uniform electron gas*, since the calculation is exact and that system is often taken as the reference problem.

Partitioning a gas of density n into non-overlapping cells of size $a = n^{-1/3}$ gives a localisation cost per cell scaling as a^{-2}, hence a kinetic energy density $t(n) = C\,n^{5/3}$ — the Thomas–Fermi form, obtained from the geometry of a partition rather than from Fermi statistics. Linearising (1) about the uniform state and substituting into Poisson's equation gives Yukawa screening with $k^2 = 18\pi n^{1/3}/5C$, and setting C to the Thomas–Fermi coefficient $C_{\mathrm{TF}} = \frac{3}{10}(3\pi^2)^{2/3} = 2.8712$ reproduces k_{TF} exactly at every density. The *form* extends without adjustment.

The coefficient does not. With the free interface used throughout RealQM, $\psi_i \equiv |\Omega_i|^{-1/2}$ is admissible: it satisfies the normalisation, makes the total density uniform, minimises the electrostatic energy against a neutralising background, and has $\nabla\psi_i = 0$. Hence $C = 0$ exactly, the screening length is infinite and the plasmon dispersionless. Neither allowing the interfaces to move nor requiring domains to be connected and singly charged alters this.

This should not be read as a test the framework fails. The uniform electron gas is electrons plus a *smeared, inert, non-responding* positive background — a construction of a tradition in which exchange–correlation energies are fitted to that very system. RealQM contains no inert backgrounds anywhere, so asking it to reproduce jellium asks it to reproduce a system whose central ingredient it does not have, and $C = 0$ diagnoses that mismatch. Consistently with this, the pathology disappears as soon as real ions are present: Table 1 is the same framework on the same physics, with the background replaced by the charges matter is actually made of.

One physical case does approach the uniform limit closely: degenerate matter in which the ion lattice is a small perturbation, as in a white dwarf interior, where the measured mass–radius relation depends on precisely the pressure the uniform limit does not supply. That is the one place the result might have observable consequences, and it is untested here.

A note on what was tested numerically

Equation (2), the screening relation, the Thomas–Fermi comparison and the vanishing of C are analytic. Table 1 is a radial Wigner–Seitz calculation with the stated pseudopotential and correlation functional, run for five alkalis; the standard column reproducing the series is what makes the comparison meaningful, and its own 44% mean error in the bulk modulus sets the accuracy available at this level of model. Two attempts are reported as null rather than omitted. An interface-condition test on helium was not converged, giving a sensitivity that reversed sign under mesh refinement, and no conclusion is drawn from it. Two chain calculations — five hydrogens with an excess electron, and five hydrogen kernels with one electron missing — returned no transport, but for a reason established by reading the solver rather than by the runs: domains are labelled by kernel proximity and bare kernels receive none, so neither hopping nor band motion is representable and neither could have been observed. Both are recorded because the null result located an obstruction that no further running would have found. Scripts and pages accompany the source.

References

[1] C. Johnson, *RealQM: Quantum Mechanics as Multiphase 3D Continuum Mechanics*, in [2].

[2] C. Johnson, *Real Quantum Mechanics* (Body&Soul, Vol. VI), `https://claes542.github.io/RealMolecule/realquantum3.pdf`.

[3] C. Johnson, *RealNucleus: Nuclear Binding as Dual Coulomb Confinement, without a Strong or Weak Force*, in [2].